From protein binders to biological behavior

Programming biological behavior with AI-guided de novo protein design

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Journal Club | Review and perspective

The central question: can designed parts compute?

The authors' thesis
Reliable molecular recognition is a foundation for engineering reaction networks whose collective behavior exceeds that of any individual binder.

1. Current strength: create a protein that recognizes a specified target.
2. Next step: couple recognition to state changes, assembly, and timed release.
3. Long-term goal: program amplification, decisions, and error correction.

This is a synthesis of other studies and a research agenda, not a new model or a single controlled benchmark.

Huettemann & Kortemme, Introduction; Sections 2–3.

Why affinity alone cannot specify a behavior

For a simple interaction $A + B \rightleftharpoons AB$,

$$K_D = \frac{[A][B]}{[AB]} = \frac{k_{\text{off}}}{k_{\text{on}}}.$$

Recognition describes

Which partner binds, how much complex accumulates, and its association and dissociation kinetics.

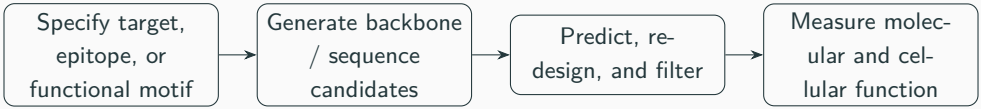
A network additionally specifies

Which event triggers the next, how signals persist, and whether an energy source creates directionality.

Two designs with the same K_D can have different residence times and therefore different signaling outputs.

Section 2.1; equation is standard mass action, included as teaching context.

A practical design workflow: proposal to cellular test



1. Diffusion constructs structures subject to target or motif constraints.
2. Inverted prediction (e.g. BindCraft) optimizes sequences using a structure predictor.
3. Sequence design and partial diffusion improve sequence–structure compatibility and fit.

Prediction confidence helps prioritize candidates; experiments establish function.

Box 1; Section 1.1. Diagram is an original synthesis.

Binder design is robust, but success is target-dependent

Example in review Table 2A	Reported binders / tested designs	Best affinity reported
RFdiffusion: IL-7 receptor	32/95 (∼34%)	30 nM
RFdiffusion: TrkA	6/95 (∼6%)	328 nM
BindCraft: cellular receptors	33/94 (∼35%)	$<$ 1–615 nM
Proteina-Complexa: PDGFR	122/192 (∼64%)	∼0.09–1,340 nM
Proteina-Complexa: ActRIIA	16/200 (8%)	36–586 nM

These are selected, heterogeneous design campaigns. Different targets, filters, and assays prevent ranking methods from this table alone.

Table 2A, review refs. 25, 27, 29. Affinity ranges follow the review's summary.

Hard recognition problems expose the missing physics

1. **Polar interfaces**: desolvation must be compensated by precise hydrogen-bond geometry. Edge-strand pairing provides a constrained solution.
2. **Flexible loops**: many conformations compete. Antibody design in the cited study required extensive screening and affinity maturation.
3. **Non-protein ligands**: small chemical groups and nucleic-acid bases demand accurate atomic interactions; training examples are scarcer.

Implication
A plausible complex structure is not enough: specificity and energetics depend on alternative states and alternative partners.

Sections 1.2 and 2.2; Table 2B.

Small molecules, DNA, and PTMs expand the input alphabet

Input	Example summarized in the review	Remaining constraint
Small molecule	RFdiffusion All-Atom: >4,000 designs tested for a 343 nM digoxigenin binder	Atomic precision and search cost
DNA sequence	RFdiffusion3 plus AlphaFold3 off-target screening: 96 designs per target	Similar global DNA geometry
Phosphotyrosine	Readers discriminate phosphorylated from unmodified peptides; high nM to low μM affinity	Affinity and selectivity in cells

Alternative small-molecule strategies predefine chemical contacts and iteratively optimize them; they trade more expert input for more efficient searches in reported examples.

Section 2.2; review refs. 26, 48–54. DNA and phosphotyrosine examples include preprints.

Disordered targets: design the bound ensemble

Modular recognition
Build repeat units recognizing successive peptide segments, then combine them into a binder.

Wu et al.: low nM binders for nearly all tested cases; 20-by-20 specificity panel showed little cross-binding.

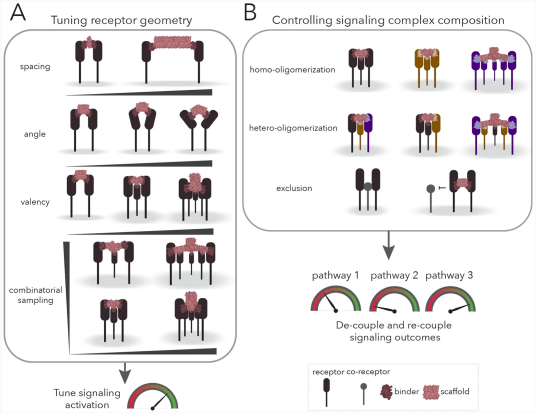
Induced structural order
Allow target and binder to move; favor a compatible helix or paired β-strand.

Flexible RFdiffusion example: specificity across eight targets and colocalization in mammalian cells.

Binding one IDR is demonstrated; programming multivalent recruitment and collective behavior is the next design problem.

Section 2.3; review refs. 16–17, 58. Evidence summarized by the review.

Assembly geometry turns binders into signaling probes



Tune distance, angle, valency, and receptor pairing independently of the native ligand.

Ask which geometry selects a downstream pathway, and which co-receptors can be excluded.

Review Figure 2; conceptual assembly diagrams, not quantitative measurements.

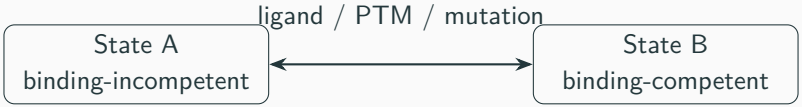
Evidence: designed assemblies rewire natural pathways

System	Perturbation / comparison	Observation summarized in review
Notch	Trimeric scaffold versus mono-/dimeric ligand	Cell clustering and T-cell differentiation; soluble agonism
TrkA	Rigid dimeric binders with different geometries	Neurite growth / differentiation with reduced inflammatory and pain-related signatures
Cytokine receptors	Change distances, rotations, and receptor combinations	Different pSTAT outputs; partial separation of inflammatory responses

Scope: these studies recruit endogenous signaling machinery. They establish useful control of natural networks, rather than fully de novo signaling pathways.

Section 2.4; refs. 60–63. TrkA and novokine studies are cited as 2025 preprints.

Dynamics: encode a landscape rather than one structure



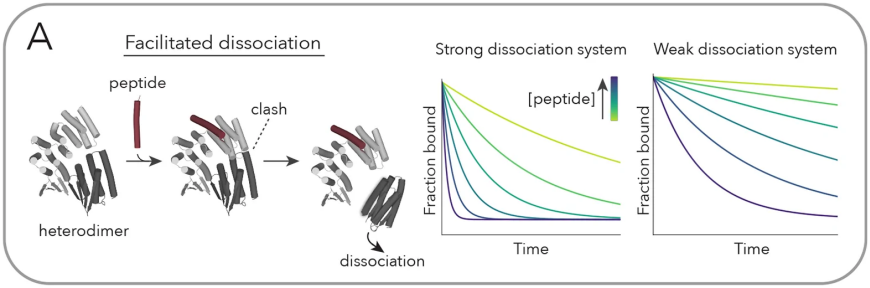
The design must control state populations, coupling to the input, and transition rates.

Guo et al. designed intradomain changes with alternative residue interaction networks. Ligands and a distal “lynchpin” mutation controlled the switch.

Limit: atomically defined switching examples do not yet supply a general recipe for arbitrary multi-step machines.

Section 2.5; ref. 65, Guo et al., Science 388, eadr7094 (2025). Diagram is a teaching schematic.

A kinetic control knob: facilitated dissociation

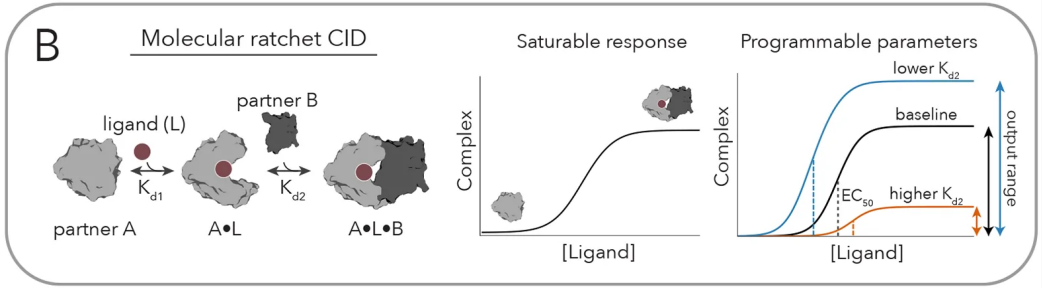


An effector peptide triggers hinge motion that ejects a bound partner. Variants accelerated dissociation by up to **2,400-fold**.

In the cited cytokine experiment, brief activation suppressed apoptosis; sustained signaling was needed for proliferation.

Figure 3A; Section 2.6; ref. 67, Broerman et al., Nature 647, 528–535 (2025). Curves are conceptual.

Topology can change the shape of a dose response

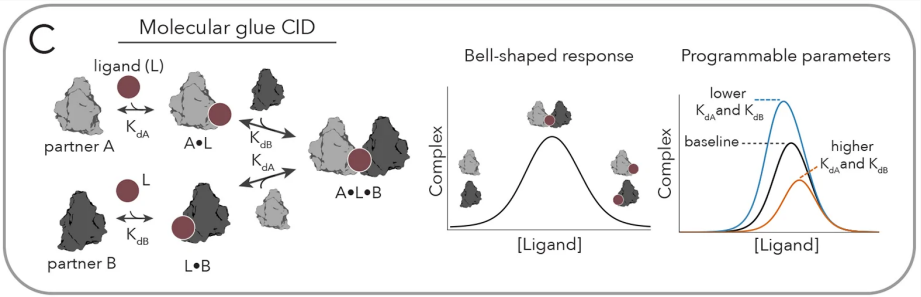


Sequential, conformationally gated CID: ligand binds A, changes its state, and permits B binding. This topology gives a saturable response in the model discussed.

Changing K_{d2} tunes sensitivity and output range. The review identifies fully programmable de novo ratchet-type CID as an outstanding goal.

Figure 3B; Section 2.6; modeling ref. 73, Steiner et al., Biochemistry 62, 281–291 (2023).

A molecular glue can behave as a band-pass filter

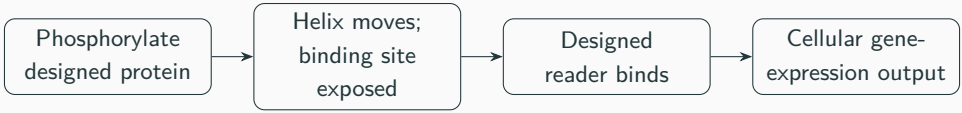


At intermediate ligand concentrations, A–ligand–B accumulates. At high ligand concentrations, separate ligands occupy A and B, reducing ternary complex formation.

The bell-shaped output follows this binding topology and parameter regime; it is not a universal property of every glue.

Figure 3C; Section 2.6; review refs. 73–74. CID = chemically induced dimerization.

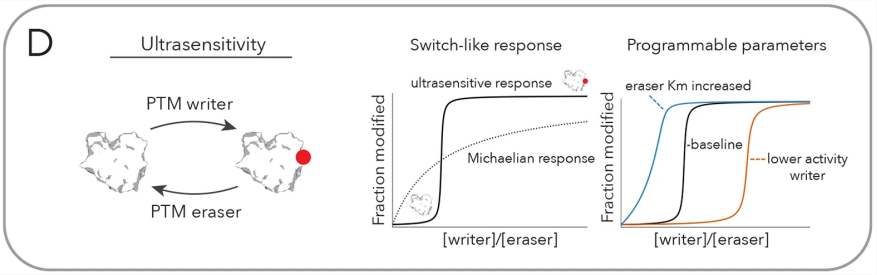
A phosphorylation switch is a component, not yet a circuit



1. Demonstrated: PTM-triggered structural change can connect to a cellular output.
2. Proposed next step: add a matched eraser and tunable writer to make a reversible regulatory circuit.
3. Stronger goal: use orthogonal PTMs and readers to reduce endogenous crosstalk.

Section 2.6; ref. 75, Buckley et al., bioRxiv (2025), DOI: 10.1101/2025.09.10.675034.

Zero-order ultrasensitivity requires opposing saturated enzymes



A writer converts S to S^*; an eraser reverses the modification. If both operate near saturation, a small change in their activity ratio can sharply shift the modified fraction.

Status in this review: established natural/theoretical principle and a design target. The phosphorylation switch alone does not establish a fully de novo ultrasensitive circuit.

Figure 3D; Section 2.1 and 2.6; refs. 41–42. Schematic curves, not new experimental data.

Proofreading needs an energy-driven reaction path



Fuel consumption

Multiple rejection opportunities can improve correct/incorrect discrimination beyond one equilibrium binding event.

The design challenge is to combine recognition, catalysis, switching, and cycling while accounting for speed and energy costs.

This is a future capability in the review's agenda, not a demonstrated de novo proofreading system.

Sections 2.1 and 3.1; review refs. 6, 43. Diagram simplifies the proofreading principle.

The scale-up opportunity has two distinct meanings

Systems-level insight

Perturb a protein's state, modification, interface, or assembly rather than deleting its gene.

The cited Human Bindome reports high-confidence computational candidates for >40% of human proteins.

Candidate coverage is not experimentally validated proteome coverage.

Systems-level engineering

Compose orthogonal components into programmable cellular circuits.

A cited synthetic heterodimer network performed winner-take-all classification in mammalian cells.

This establishes an example of computation, not arbitrary circuit composability.

Sections 3.2–3.3; refs. 77 (Human Bindome, 2026 preprint), 78 (Chen et al., Science, 2024).

What would convince us that the parts are composable?

Our discussion framework, motivated by the review:

1. **Molecular test**: measure affinity, specificity, rates, and state populations with matched controls.
2. **Circuit test**: predict a dose/time response from independently measured component parameters, then test the prediction.
3. **Context test**: vary expression and cellular background; quantify leakage, crosstalk, burden, and failure modes.

A high-affinity binder establishes recognition. A predictable transfer function across perturbations establishes a stronger engineering capability.

Interpretation / proposed evaluation framework; relates to Sections 2.6 and 3.4, not an experiment in this review.

Discussion: what should be designed and measured next?

1. Which minimal circuit would most clearly demonstrate programmable behavior: reversible PTM switching, timed receptor activation, or an orthogonal assembly gate?
2. Which missing measurement is most limiting: rate constants, conformational landscapes, energy coupling, or cellular off-target effects?
3. Can independent validation break the circularity of using related prediction models for both generation and selection?

Take-home
Recognition is increasingly practical. Predictable biological behavior requires quantitative control of states, rates, topology, and cellular context; energy-driven circuits remain a major frontier.

Synthesis of the review; questions are journal-club discussion prompts.

Sources and reading guide

Main source

Huettemann P, Kortemme T. *Molecular Cell* 86(20), 4167–4182 (15 October 2026). DOI: 10.1016/j.molcel.2026.08.027.

`https://pmc.ncbi.nlm.nih.gov/articles/PMC13679063/`

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Figures 2 and 3 are reproduced from the accessible author manuscript. Figure 3 is a set of mechanistic illustrations, not a common experimental benchmark. Quantitative examples and primary-paper references are attributed to the review.

Access limitation

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